Index laws

1 Write the following expressions in simplest index form:

a $2^{12} \times 2^{9}$

b $2^{8} \times 11^{8}$

c $11^{12} \div 11^{8}$

d $21^{5} \div 3^{5}$

e $\left(5^{12}\right)^{4}$

f $15^{17} \div 15^{8} \div 15^{5}$

g $\left(23^{8}\right)^{9} \times 23^{7}$

h $\dfrac{\left(17^{5}\right)^{8}}{17^{32}}$

i $\dfrac{19^{9} \times 19^{4}}{19^{8}}$

j $\dfrac{12^{6}}{12^{4}} \times 12^{5}$

k $\dfrac{\left(13^{5}\right)^{2} \times 13^{3}}{13^{5}}$

l $\dfrac{\left(15^{9}\right)^{5} \times 15^{7}}{15^{25}}$

2 Complete the following statements:

a $11^{11} \times 11^{\square} = 11^{19}$

b $5^{11} \times (\square)^{11} = 35^{11}$

c $97^{\square} \div 97^{22} = 97^{12}$

d $55^{6} \div (\square)^{6} = 11^{6}$

e $\left(11^{4}\right)^{\square} = 11^{12}$

f $7^{8} \times 7^{\square} = 7^{14}$

g $2^{10} \times (\square)^{10} = 10^{10}$

h $19^{\square} \div 19^{18} = 19^{20}$

i $60^{4} \div (\square)^{4} = 12^{4}$

j $\left(13^{8}\right)^{\square} = 13^{16}$

3 Evaluate the following expressions:

a $6^{5} \times 6^{3}$

b $7^{3} \times 3^{3}$

c $4^{8} \div 4^{3}$

d $\left(5^{4}\right)^{2}$

e $35^{5} \div 5^{5}$

f $2^{4} \times 4^{4}$

g $11^{18} \div 11^{9} \div 11^{7}$

h $\left(3^{3}\right)^{2}$

i $\dfrac{6^{5} \times 6^{9}}{6^{12}}$

j $7^{27} \div 7^{30} \div 7^{3}$

k $\dfrac{12^{10} \times 12^{4}}{12^{11}}$

l $\dfrac{\left(6^{8}\right)^{6}}{6^{46}}$

Negative bases

4 Write the following expressions in simplest index form:

a $(-11)^{10} \times (-11)^{3}$

b $(-7)^{8} \times 3^{8}$

c $(-5)^{2} \times 3^{2}$

d $(-3)^{12} \div (-3)^{5}$

e $(-12)^{20} \div (-12)^{19}$

f $(-30)^{50} \div (-30)^{47}$

g $(-48)^{3} \div (-6)^{3}$

h $(-33)^{11} \div (-3)^{11}$

i $(-35)^{5} \div 5^{5}$

j $(-42)^{2} \div 7^{2}$

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5 Complete the following statements:

a $11^3 \times (\square)^3 = (-77)^3$

b $(-5)^{11} \times (\square)^{11} = 15^{11}$

c $(-5)^{\square} \div (-5)^{31} = (-5)^{19}$

d $(-14)^{13} \div (\square)^{13} = (-7)^{13}$

e $(-3)^{\square} \div (-3)^{39} = (-3)^{10}$

f $(-5)^4 \times (\square)^4 = (-60)^4$

g $(-3)^7 \times (\square)^7 = (6)^7$

h $(\square)^3 \div (-3)^3 = (7)^3$

i $(-33)^3 \div (\square)^3 = -11^3$

j $(\square)^7 \div (-5)^7 = 11^7$

6 Evaluate the following expressions:

a $(-4)^{11} \div (-4)^7$

b $(-2)^3 \times (-2)^3$

c $(-3)^3 \times (-3)^2$

d $4^3 \times (-5)^3$

e $(-3)^8 \div (-3)^5$

f $15^5 \div (-3)^5$

g $2^3 \times (-3)^3$

h $(-14)^{11} \div 2^{11}$

i $(-7)^2 \times 5^2$

j $(-9)^4 \times (-3)^4$

k $(-100)^6 \div 50^6$

l $60^3 \div (-3)^3$

Fractional bases

7 Write the following in simplest index form:

a $\left(\frac{1}{3}\right)^4$

b $\left(\frac{3}{8}\right)^3$

c $\left(\frac{4}{16}\right)^8$

d $\left(\frac{15}{6}\right)^2$

e $\left(\frac{10}{33}\right)^5$

f $\left(\frac{2}{35}\right)^6$

g $\left(\frac{5}{18}\right)^3$

h $\left(\frac{29}{41}\right)^7$

i $\left(\frac{11}{13}\right)^9$

j $\left(\frac{20}{3}\right)^2$

k $\left(\frac{17}{4}\right)^4$

l $\left(\frac{31}{50}\right)^5$

8 Complete the following statements:

a $\frac{1}{27} = \left(\frac{1}{3}\right)^{\square}$

b $\frac{64}{27} = \left(\frac{4}{3}\right)^{\square}$

c $\frac{27}{8} = \left(\frac{3}{\square}\right)^3$

d $\frac{\square}{16} = \left(\frac{1}{4}\right)^2$

e $\frac{27}{8} = \left(\frac{3}{2}\right)^{\square}$

f $\frac{\square}{625} = \left(\frac{2}{5}\right)^4$

g $\frac{81}{100} = \left(\frac{\square}{10}\right)^2$

h $\frac{256}{\square} = \left(\frac{4}{5}\right)^4$

i $\frac{25}{9} = \left(\frac{5}{3}\right)^{\square}$

j $\frac{\square}{144} = \left(\frac{11}{12}\right)^2$

k $\frac{125}{27} = \left(\frac{\square}{3}\right)^3$

l $\frac{32}{729} = \left(\frac{2}{3}\right)^{\square}$

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9 Evaluate the following expressions in fully simplified fraction:

a $\left(\frac{1}{3}\right)^2$ b $\left(\frac{3}{5}\right)^3$ c $\left(\frac{1}{2}\right)^5$ d $\left(\frac{7}{11}\right)^2$

e $\left(\frac{3}{8}\right)^3$ f $\left(\frac{12}{13}\right)^2$ g $\left(\frac{5}{9}\right)^3$ h $\left(\frac{2}{3}\right)^4$

i $\left(\frac{4}{5}\right)^6$ j $\left(\frac{2}{5}\right)^3$ k $\left(\frac{5}{7}\right)^2$ l $\left(\frac{10}{14}\right)^3$

10 State whether the following fractions are equal to $\left(\frac{1}{2}\right)^3$:

a $\frac{3}{6}$ b $\frac{1}{2^3}$ c $\frac{3}{2^3}$ d $\frac{1}{8}$

Negative indices

11 Is 2^{-3} less than or greater than 1?

12 Express the following with positive indices:

a a^{-9} b $\frac{1}{a^{-n}}$ c $\frac{a^{-9}}{4}$ d $\frac{a^{-n}}{b^{-m}}$

e p^{-2} f $3x^{-4}$ g $7x^{-9}$ h $p^{-2}q^3$

i $8p^{-3}$ j $2x^{-8}y^3$ k $\frac{b^{-7}}{6c^{-3}}$ l $m^{-5}n^{-4}p^4$

m $3a^4b^{-5}c^{-7}$ n $4^{-2}k^{-3}l^7$ o $\frac{f^{-7}}{g^{-2}h^8}$ p $\frac{s^4t^{-9}}{5u^{-6}}$

13 Express the following without fractions:

a $\frac{1}{u^4}$ b $\frac{2}{r^6}$ c $\frac{x}{y^2}$ d $\frac{4}{p^6q^7}$

14 Simplify the following, giving your answers with positive indices:

a $5y^9 \times 4y^{-3}$ b $7a^4 \times 4a^{-2}$ c $5x^4 \times (-3x^{-8})$ d $3y^{-2} \times 4y^{-3}$

e $2h^{-4} \times 4h^{11}$ f $3y^{-2} \times 2y^{-5}$ g $-4y^2 \times (-4y^{-5})$ h $(5mp)^2 \times mp^{-2}$

i $\frac{9x^2}{3x^9}$ j $\frac{15x^3}{5x^7}$ k $12x^5 \div 4x^{-3}$ l $9c^6 \div 6c^8$

m $\frac{15h^2}{12h^{-7}}$ n $\frac{12x^5}{4x^{-3}}$ o $\frac{21p^{-4}}{14p^{-3}}$ p $\frac{36w^{-6}}{16w^4}$

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15 Simplify the following, giving your answers with positive indices:

a $(2m)^{-3}$ **b** $\left(4m^{-6}\right)^{4}$ **c** $\left(3p^{-4}\right)^{-2}$ **d** $\left(3y^{2}\right)^{-2}$

e $\left(\frac{y}{4}\right)^{-3}$ **f** $\left(\frac{x^{5}}{y^{4}}\right)^{-1}$ **g** $\left(\frac{x^{7}}{y^{9}}\right)^{-4}$ **h** $\frac{20x^{3}}{4x^{-2}}$

i $\frac{10x^{-7}}{2x^{-3}}$ **j** $\left(\frac{z}{3}\right)^{-4}$ **k** $\left(\frac{p^{3}}{q^{7}}\right)^{-1}$ **l** $\left(\frac{x^{-4}}{y^{-8}}\right)^{-2}$

16 Solve the following equations for n:

a $\frac{1}{25}=5^{n}$ **b** $\frac{1}{8}=2^{n}$

c $\left(x^{3}y^{-5}\right)^{n}=x^{-12}y^{20}$ **d** $\left(a^{-5}b^{3}\right)^{n}=a^{15}b^{-9}$

17 Complete the following:

$$\left(x^{4}y^{\blacksquare}\right)^{\blacksquare}=\frac{y^{12}}{x^{16}}$$

18 A student was writing $5a^{-1}$ without negative indices and wrote $5a^{-1}=\frac{1}{5a}$. Explain why their working is incorrect, and write the correct answer.

19 Simplify:

a $2y^{6}\times 4y^{7}\times 4y^{-5}$ **b** $6y^{7}\times 2y^{-5}\times 5y^{3}$

20 Simplify the following, giving your answers with positive indices:

a $\frac{2x^{2}y^{-4}}{7z^{-3}}\times\frac{5x^{-5}z^{6}}{3y^{3}}$ **b** $\left(\frac{3x^{-2}}{10y^{4}z^{-1}}\right)^{-2}\div\left(\frac{8x^{-5}z^{2}}{15y^{2}}\right)^{-1}$

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